Lecture 9: Financial Derivatives, Hedging And Option Payoffs

Phase 2: Risk Management Science

OSALP 2025-04
Nurturing Future Risk Managers and Actuaries

Department of Statistics and Data Science
The Chinese University of Hong Kong

Gifted Education Fund Programme

2026

⋆ Core What This Revised Lecture Focuses On I

Key Definition / Result

The central theme is now:

How can contracts reshape risk without changing the underlying asset itself?

1. Local derivative language: margin trading, warrants, and callable bull/bear contracts.
2. Underlying assets and risk-free discounting: only the background needed for derivative contracts.
3. Forwards and futures: locking future prices, hedging, counterparty risk, clearing houses, and margin risk.
4. Leverage: why derivatives can magnify both gains and losses.
5. Options: call options, put options, payoff diagrams, profit diagrams, premium, and exercise choice.
6. No-arbitrage thinking: put-call parity and replication intuition.

★ Core

What This Revised Lecture Focuses On II

7. Extensions: cross-hedging and Black–Scholes–Merton as optional self-directed explorations.

Common Mistake / Pitfall

This deck no longer teaches portfolio covariance, Markowitz theory, or efficient frontiers. Those belong to a separate market-risk or portfolio-management lecture.

★★ Insight Do You Know? Local Financial Products I

In Hong Kong's financial market, students often encounter these terms in daily news and retail investing discussions:

- **F展 (Margin Trading):** leveraged trading using borrowed funds to purchase securities.
- **窩輪 (Warrants / Options):** derivative instruments issued by financial institutions, giving the holder structured rights tied to an underlying asset.
- **牛熊證 (Callable Bull/Bear Contracts, CBBC):** structured products with built-in knock-out features.

Cognitive Trigger / Opening Problem

Why are these retail derivatives so popular in Hong Kong? How do they differ in risk from holding the underlying stocks directly?

Pause-and-Think

If a product can multiply gains, what hidden mechanism may also multiply losses?

Opening Problem: Locking In A Future Price I

Cognitive Trigger / Opening Problem

A student is planning a winter trip to Japan in December, but today the JPY/HKD exchange rate looks attractive. At the same time, a contractor worries that metal prices may rise before a construction project begins.

Questions

1. How can each person lock in a future price today?
2. Why can such contracts both reduce risk and magnify gains or losses?
3. Why do derivatives exist even when the underlying assets already exist?

Roadmap: One Derivatives Story I

1. Part A. Underlying assets and discounting: the minimum background needed for derivatives.
2. Part B. Forwards and futures: agreement, payoff, hedging, clearing, and margin.
3. Part C. Leverage: why small price movements can create large percentage gains or losses.
4. Part D. Options: asymmetric rights, premiums, payoff diagrams, and profit diagrams.
5. Part E. Put-call parity: replication and no-arbitrage as a first structural pricing idea.
6. Part F. Investigation and review: diagrams, interpretation, and challenge questions.

Technology Check: Learning Design

This lesson is intentionally payoff-first and interpretation-first. Formal stochastic calculus, continuous-time hedging, and advanced option pricing are reserved for optional self-directed exploration.

Part A. Underlying Assets And Discounting Background

⋆ Core

Underlying Assets: What Can A Derivative Be Linked To? I

Key Definition / Result

A derivative is meaningful only because there is an underlying asset or underlying variable.

- Stocks: partial ownership of a company.
- Commodities: gold, oil, rice, metals, or other traded goods.
- Foreign currencies: USD, JPY, CNY, HKD, and exchange rates.
- Bonds or interest rates: debt instruments and discount rates.
- Indices or baskets: market indices or groups of securities.

Investigation / Activity

A derivative contract does not usually create the underlying asset. Instead, it creates a new payoff rule linked to that asset.

⋆ Core Stocks And Bonds: Only The Background We Need I

Key Definition / Result: Stocks

Stocks represent ownership of a company. Their prices are uncertain and depend on future expectations, earnings, market sentiment, and economic conditions.

Key Definition / Result: Bonds

Bonds represent lending money to an issuer. Their value depends on promised payments, discounting, interest rates, and credit risk.

Technology Check

For this derivatives lecture, stocks mainly act as risky underlying assets, while risk-free bonds mainly help us understand discounting and put-call parity.

Risk-Free Discounting: Why A Bond Appears In Option Pricing I

Key Definition / Result

For a risk-free annual effective interest rate r,

$$B_T = B_0(1+r)^T, \qquad B_0 = \frac{B_T}{(1+r)^T}.$$

Worked Example

Suppose $B_0 = 100$, $r = 3\%$, and $T = 10$. Then

$$\text{Interest} = B_T - B_0 = 100\left((1+0.03)^{10} - 1\right) = \textbf{34.392}.$$

Investigation / Activity

This discounting idea will reappear later when we compare an option portfolio with a stock-plus-bond portfolio.

★★ Insight

Quick Reminder: Promised Payment Is Not The Same As Certain Payment I

Worked Example

A 1-year bond pays coupon $\$50$ and face value $\$1000$. If the risk-free rate is 5%, then for a default-free bond,

$$P = \frac{50 + 1000}{1 + 0.05} = \mathbf{\$1000.00}.$$

If the same promised payoff has 10% default probability and zero recovery,

$$E[\text{Payoff}] = 0.90 \times 1050 + 0.10 \times 0 = 945,$$

so

$$P_{\text{risky}} = \frac{945}{1.05} = \mathbf{\$900.00}.$$

★★ Insight Quick Reminder: Promised Payment Is Not The Same As Certain Payment II

Common Mistake / Pitfall

A large promised payment does not automatically mean a high value today. Probability of payment matters.

Part B. Forwards And Futures

Financial Derivatives: A First Definition I

Key Definition / Result

A financial derivative is a contract whose value depends on the future outcome of an underlying asset or variable.

- Forwards / Futures (期貨): agreements to trade an item on a future date at a pre-determined price.
- Options (期權 / 窩輪): rights to trade an item on a future date at a pre-determined price, where the holder may choose whether to exercise.

Proof / Modelling Decision Point

Derivatives are tools that can be used to eliminate, transfer, reshape, or trade risk. The product is not automatically good or bad; the use case matters.

Hedging Exchange-Rate Risk: JPY/HKD Scenario I

Worked Example

A student plans a winter trip to Japan in December.

- The current exchange rate looks attractive.
- The student does not have money to exchange yen right now.
- The risk is that JPY appreciates by December, making the trip more expensive.
- A possible solution is an exchange-rate forward to exchange 100000 JPY with 7000 HKD in December.

Investigation / Activity

This is a hedging example: the student gives up exchange-rate uncertainty by agreeing to a future exchange rate today.

★★ Insight

Hedging Exchange-Rate Risk: Trade-Off Analysis I

Working Example: Hedge Protects

If JPY appreciates in December so that

$$100000\,\text{JPY} = 8000\,\text{HKD},$$

the student still pays $7000\,\text{HKD}$, so the hedge protects the trip budget.

Boundary / Failing Example: Hedge Has Opportunity Cost

If JPY depreciates in December so that

$$100000\,\text{JPY} = 6000\,\text{HKD},$$

the student is still legally obligated to pay $7000\,\text{HKD}$, so the student gives up upside.

Pause-and-Think

Why does a forward eliminate upside potential while locking in downside protection? How might an option solve this limitation, and at what cost?

Forward Contract: Private Agreement To Trade Later I

Key Definition / Result

A forward contract is an agreement today to buy or sell an underlying asset at a specified future time for a specified price K.

- The buyer of the forward agrees to buy the asset at price K.
- The seller of the forward agrees to sell the asset at price K.
- The payoff depends on the future spot price S_T.

Worked Example: Keith And Bosco's Car Deal

Keith is leaving Hong Kong next month. Bosco agrees today to buy Keith's car for $\$2000$ next month. This is a forward contract, and the forward price is $\$2000$.

⋆ Core Futures Contract: Standardised Exchange-Traded Forward-Like Contract I

Key Definition / Result

A futures contract has similar economic logic to a forward contract, but it is standardised and typically traded through an exchange with margin and regular settlement.

Worked Example: Metal Price Hedge

A contractor cannot afford a hike in metal price. The contractor goes **long metal futures** at futures price K. This guarantees a purchase price K in the future, while performance is supported by a clearing house.

Pause-and-Think

How might a clearing house guarantee settlement even if one side goes bankrupt?

★ Core

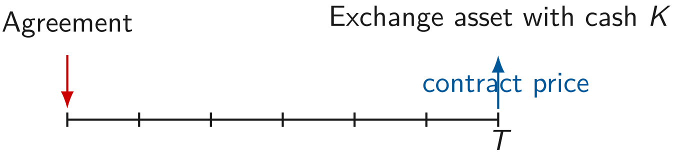
Forwards Versus Futures: A Structural Contrast I

Dimension	Forward Contracts	Futures Contracts
Trading venue	OTC private deal	Organised exchange, such as HKEx
Standardisation	Highly customised	Highly standardised
Credit risk	Counterparty risk matters	Clearing-house protection
Settlement	Usually one payment at maturity	Daily mark-to-market / margin
Example	Keith and Bosco's car deal	Standardised metal futures

Investigation / Activity

Inquiry task: how does contract standardisation help liquidity and exchange trading?

Forward And Futures Timeline I



Key Definition / Result

The key modelling idea is that the price is fixed in advance, but the future market price S_T is uncertain.

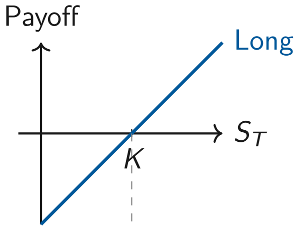
Futures Payoff: The Long Position I

Key Definition / Result

The payoff at maturity T is linear:

$$\text{Payoff}_{\text{Long}} = S_T - K.$$

- If $S_T > K$, the payoff is positive.
- If $S_T < K$, the payoff is negative.
- There is large upside and also large downside.



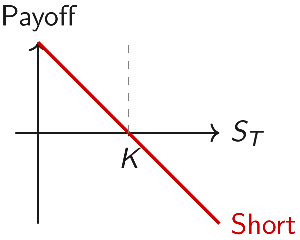
⋆ Core Futures Payoff: The Short Position I

Key Definition / Result

The payoff at maturity T is linear:

$$\text{Payoff}_{\text{Short}} = K - S_T.$$

- If $S_T < K$, the payoff is positive.
- If $S_T > K$, the payoff is negative.
- There is significant upside and potentially unlimited downside.



Forwards And Futures: Contract Price Versus Contract Value I

Key Definition / Result

At time $t = 0$, the value of entering a fair futures or forward-style contract is often treated as zero. As time evolves, the value of the position becomes non-zero when the market price changes.

Common Mistake / Pitfall

Do not confuse the agreed contract price K with the value of the contract after market prices have moved.

★★ Insight

Forwards Versus Futures: Counterparty And Liquidity Risk I

Common Mistake / Pitfall

Futures do not eliminate all risk. Futures reduce credit or counterparty risk through the clearing house, but they introduce liquidity or margin-call risk because losses are settled regularly. If an investor lacks cash to meet a margin call, the position may be liquidated.

Investigation / Activity

Research prompt: read about Metallgesellschaft in 1993. How can a short-term margin shortfall destroy a structurally sensible long-term hedge?

Part C. Leverage And Margin Risk

⋆ Core Leverage: Small Capital, Large Exposure I

Key Definition / Result

Leverage means controlling a large economic exposure with a smaller amount of capital.

- A direct stock purchase requires paying the full stock price.
- A futures-style position may require only margin or a smaller initial commitment.
- The same underlying price movement can therefore become a much larger percentage gain or loss on the investor's own capital.

Common Mistake / Pitfall

Leverage is not free money. It changes the relationship between capital, exposure, gain, loss, and margin pressure.

★★ Insight

Leverage Property Of Futures: Good Market I

Worked Example

Futures can turn a good investment into a great investment.

Your capital: \$10000; today's stock price: \$10.0; risk-free rate: 10%; futures price: \$8.8; value of futures contract: \$2.	
All in stock Units of stock: $10000/10 = 1000$ If stock price rises from 10 to 12: Profit: $(12 - 10) \times 1000 =$ **2000**	**All in future** Units of future: $10000/2 = 5000$ If stock price rises from 10 to 12: $(12 - 8.8) \times 5000 - 10000 =$ **6000**

The Leverage Property: Scenario A, The Upside I

Worked Example

Suppose $S_0 = 100$, futures price $F_0 = 100$, and margin $= 10$. If the stock price rises to 110, the futures payoff is 10. Hence,

$$\text{Return} = \frac{10}{10} = \mathbf{+100\%}.$$

Pause-and-Think

How does the clearing house protect itself if the winning side gains $+100\%$ but the losing side cannot pay?

★★ Insight

Leverage Property Of Futures: Bad Market I

Worked Example

Futures can also turn a bad investment into a terrible investment.

Your capital: $\$10000$; today's stock price: $\$10.0$; risk-free rate: 10%; futures price: $\$8.8$; value of futures contract: $\$2$.	
All in stock Units of stock: $10000/10 = 1000$ If stock price drops from 10 to 8: Loss: $2 \times 1000 =$ **2000**	**All in future** Units of future: $10000/2 = 5000$ If stock price drops from 10 to 8: $(8 - 8.8) \times 5000 - 10000 =$ **-14000**

The Leverage Property: Scenario B, The Downside I

Worked Example

Suppose $S_0 = 100$, futures price $F_0 = 100$, and margin $= 10$. If the stock price falls to 90, the futures payoff is -10. Hence,

$$\text{Return} = \frac{-10}{10} = \textbf{-100\%}.$$

Pause-and-Think

If the stock falls further to 85, what happens if the investor fails to deposit more cash?

⋆ Core

Leverage Summary: What Students Must Remember I

Key Definition / Result

Leverage changes the scale of exposure relative to the student's own capital.

Working Example: Useful Use

A hedger may use futures to reduce uncertainty about a future purchase or sale.

Boundary / Failing Example: Dangerous Use

A speculator may use futures to take a large position with little capital, creating losses larger than the original capital.

Common Mistake / Pitfall

A futures hedge can be sensible economically but still fail operationally if the hedger cannot survive margin calls.

Multi-Asset Futures Hedging And Cross-Hedging I

Investigation / Activity

How do corporations hedge risk when there is no standardised futures contract on the exact asset they hold? They may use cross-hedging.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

`Explain how cross-hedging works when there is no direct futures contract on the underlying asset, such as hedging jet fuel using crude oil futures. Explain the intuition behind the optimal hedge ratio` $h^* = \rho\sigma_S/\sigma_F$`, and provide an R script to evaluate hedging effectiveness.`

Part D. Options And Payoff Diagrams

⋆ Core Options: Asymmetric Rights I

Key Definition / Result

There are two basic types of options:

- Call options: the right to buy an asset for strike price K at maturity T.
- Put options: the right to sell an asset for strike price K at maturity T.

$$\text{Payoff}_{\text{Call}} = \max(S_T - K, 0), \qquad \text{Payoff}_{\text{Put}} = \max(K - S_T, 0).$$

Technology Check

The holder may choose whether or not to exercise this right. That optionality is why the option has a premium.

Call Option Timeline I

A call option involves paying an upfront premium c at $t = 0$ for a future right.

Pause-and-Think

Why must the premium c be non-negative today even if the future payoff may turn out to be zero?

★★ Insight

Put Option Timeline I

A put option involves paying an upfront premium p at $t = 0$ for a future right to sell.

Pause-and-Think

Under what market conditions does a put become especially valuable to risk managers?

Worked Example: Future Versus Option I

Worked Example: Future

At $t = 0$, you enter a futures contract to buy stock S for future price $f = 50$ with maturity T. At T, the stock price is $S_T = 45$. To fulfil the futures contract, you acquire stock S at cost 50. Your payoff is

-5.

Worked Example: Call Option

At $t = 0$, you enter a call option contract on stock S with strike price $f = 50$ and maturity T.

- If $S_T = 45$, you do not exercise, so the payoff is **0**.
- If $S_T = 55$, you exercise, so the payoff is **+5**.

Options: Four Basic Payoffs I

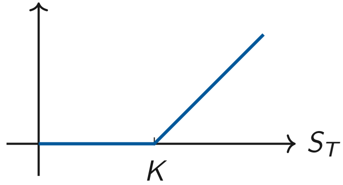
Option Position	Payoff
Long position in a call option	$+\max(S_T - K, 0)$
Long position in a put option	$+\max(K - S_T, 0)$
Short position in a call option	$-\max(S_T - K, 0)$
Short position in a put option	$-\max(K - S_T, 0)$

Common Mistake / Pitfall

A payoff diagram usually ignores the upfront premium. A profit diagram includes the premium.

Call Options: Payoff At Maturity I

Payoff at maturity

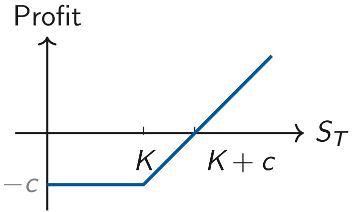


Key Definition / Result

A long call has zero payoff below the strike and increasing payoff above the strike.

Call Options: Profit With Premium I

Profit with premium c

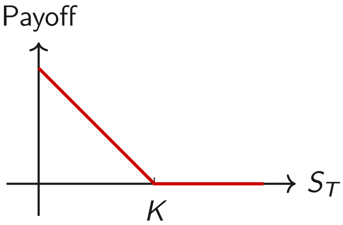


Technology Check

The break-even stock price at maturity is $S_T = K + c$. Below K, the maximum loss is exactly $-c$.

★★ Insight Put Options: Payoff At Maturity I

Payoff at maturity



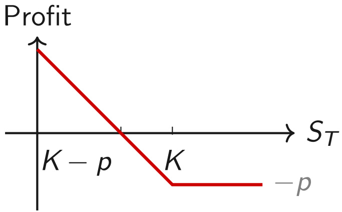
Key Definition / Result

A long put has increasing payoff when the underlying price falls below the strike.

★★ Insight

Put Options: Profit With Premium I

Profit with premium p



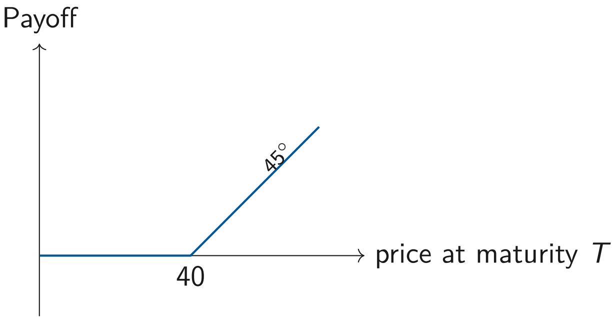
Technology Check

The break-even stock price at maturity is $S_T = K - p$. Above K, the maximum loss is exactly $-p$.

Options: Payoff Diagram For A Long Call I

Investigation / Activity

Draw the payoff diagram of a long call with strike $K = 40$ and call price $c = 2$. Distinguish carefully between payoff and profit.

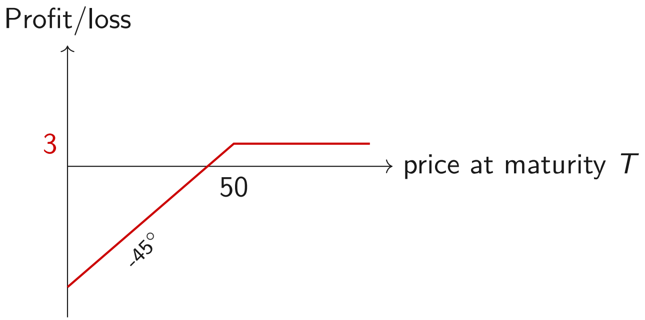


★★ Insight

Options: Profit/Loss Diagram For A Short Put I

Investigation / Activity

Draw the profit/loss diagram of a short put with strike $K = 50$ and put price $p = 3$.



⋆⋆⋆ Extension

Exercise: Building Option Diagrams I

Investigation / Activity

1. Draw the payoff diagram of 2 long calls with strike $K = 25$ and call price $c = 15$.
2. Draw the profit/loss diagram of a short put and a long call, both with strike $K = 50$, put price $p = 3$, and call price $c = 2$.

Technology Check

For each diagram, label the strike, premium, break-even point, maximum loss, and region where the position makes money.

★ Core

Forward, Future, Call And Put: Side-By-Side I

Instrument	Holder's position	Payoff structure	Risk-management role
Forward	Obligation to trade later	Linear payoff	Lock in a future price
Future	Exchange-traded obligation with margin	Linear payoff with margin settlement	Hedge or speculate with clearing-house support
Call option	Right to buy	$\max(S_T - K, 0)$	Preserve upside exposure
Put option	Right to sell	$\max(K - S_T, 0)$	Protect against downside loss

Common Mistake / Pitfall

The key distinction is obligation versus right. A forward or future binds both sides; an option gives the holder a choice.

Part E. Put–Call Parity And No-Arbitrage Thinking

Proof Decision Point: Put–Call Parity I

Key Definition / Result

For European options on non-dividend-paying stocks:

$$c + Ke^{-rT} = p + S_0.$$

- Portfolio A: one call plus a zero-coupon bond paying K at maturity.
- Portfolio B: one put plus one share of stock.
- At maturity, both portfolios have value $\max(S_T, K)$.
- Under no-arbitrage, if two portfolios have identical future payoffs in all states, they must have identical initial costs.

Why Put–Call Parity Is A Replication Idea I

Future state	Call + bond	Put + stock
$S_T > K$	Call pays $S_T - K$, bond pays K, total S_T	Put pays 0, stock value S_T, total S_T
$S_T \leq K$	Call pays 0, bond pays K, total K	Put pays $K - S_T$, stock value S_T, total K

Key Definition / Result

The two portfolios match in every future state. Therefore, under no-arbitrage, they should have the same cost today.

Put–Call Parity Arbitrage: Case 1 I

Worked Example

Suppose $S_0 = \$100$, $K = \$100$, $r = 0\%$, and $T = 1$. If the market prices are $c = \$15$ and $p = \$10$, then

$$c + K = 15 + 100 = 115,$$
$$p + S_0 = 10 + 100 = 110.$$

Since $115 > 110$, the call side is overpriced.
Selling the call, buying the put, buying the stock, and borrowing $\$90$ locks in an immediate arbitrage profit of **$5.00**.

★★ Insight

Put–Call Parity Arbitrage: Case 2 I

Worked Example

If $c = \$12$, $p = \$12$, $S_0 = \$100$, $K = \$100$, and $r = 0\%$, then

$$c + K = 12 + 100 = 112,$$

$$p + S_0 = 12 + 100 = 112.$$

The parity holds, so there is no arbitrage profit to extract.

Pause-and-Think

What changes when the interest rate $r > 0$?

⋆ Core Why No-Arbitrage Matters For Risk Management I

Key Definition / Result

No-arbitrage thinking says that two portfolios with the same future payoff in every state should not have different prices today.

- It helps detect inconsistent market prices.
- It provides a logical foundation for derivative pricing.
- It shifts attention from predicting one future price to comparing payoff structures across all possible states.

Common Mistake / Pitfall

No-arbitrage does not mean everyone can always get free money in real markets. It is a modelling principle that explains why large price inconsistencies should be competed away.

⋆⋆⋆ Extension Extension Layer: The Black–Scholes–Merton PDE I

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

`Explain the replication argument of the Black-Scholes-Merton model using a self-financing portfolio of a stock and a risk-free bond. Investigate the mathematics behind continuous-time hedging using calculus concepts outside of class time. Discover how it leads to continuous option pricing, and explain Delta, Gamma, and Vega in plain language.`

Part F. Investigation And Review

Investigation: Hedging Or Speculation? I

Investigation / Activity

For each situation, decide whether the derivative position is mainly hedging, speculation, or ambiguous.

1. A traveller locks in a future JPY/HKD exchange rate before a trip.
2. A contractor buys metal futures before purchasing metal.
3. A student buys a call option because they believe a stock will rise.
4. A shareholder buys a put option to protect against a market crash.
5. A trader sells many put options to collect premium.

Technology Check

For each case, write one sentence explaining the underlying exposure and one sentence explaining the derivative payoff.

Investigation: Build A Payoff Diagram Gallery I

Investigation / Activity

Students create a four-panel payoff diagram gallery:

1. long forward;
2. short forward;
3. long call;
4. long put.

Then they add a second version that includes premium for the option diagrams.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

`Explain how to draw payoff and profit diagrams for a long forward, short forward, long call, and long put. Use one common strike or contract price` $K = 50$`, and explain the difference between payoff and profit.`

Review Set 9A: Easier Practice, Part 1 I

1. Define a derivative in one sentence.
2. Explain the difference between a forward and a future.
3. Write the payoff formula for a long futures position.
4. Write the payoff formula for a short futures position.
5. Explain why a futures contract can be used for hedging.

Technology Check

For every formula, write one interpretation sentence. A correct formula without interpretation is not yet a risk-management answer.

Review Set 9A: Easier Practice, Part 2 I

6. State the difference between a call option and a put option.
7. Compute the payoff of a long call with $K = 30$ if $S_T = 42$.
8. Compute the payoff of a long put with $K = 30$ if $S_T = 24$.
9. Compute the payoff of a short put with $K = 20$ if $S_T = 16$.
10. Explain why option premium changes payoff into profit.

Review Set 9B: More Challenging Practice, Part 1 I

1. Draw the payoff diagram of a short futures contract with price K.
2. Find the profit/loss diagram of a long call purchased for premium c.
3. Explain in context why leverage can be useful but dangerous.
4. Give one example where a futures hedge may fail because of margin-call risk.

Review Set 9B: More Challenging Practice, Part 2 I

5. Explain why a forward contract removes both downside risk and upside opportunity.
6. Compare buying a put option with taking a short futures position as downside protection.
7. Explain put–call parity as a replication or no-arbitrage idea in plain language.
8. Describe one reason a derivative product may be popular even when it is risky.

Review Problems: Challenge Your Thinking I

Investigation / Activity: Set A: Core

Under what circumstances would an investor choose a short futures position instead of buying a put option, given that the put limits maximum loss while the futures contract does not?

Investigation / Activity: Set B: Insight

Why might a product that is useful for hedging become dangerous when used for speculation?

Investigation / Activity: Set C: Extension

How might transaction costs, bid–ask spreads, liquidity risk, and margin calls weaken the clean textbook idea of no-arbitrage?

Coverage Note For This Derivatives-Focused Version I

Key Definition / Result

This deck keeps the original Lecture 9 derivatives content: local Hong Kong products, underlying instruments, bond discounting background, forwards, futures, hedging, leverage, options, payoff diagrams, put-call parity, arbitrage intuition, and optional Black–Scholes–Merton exploration.

Technology Check

This deck deliberately removes the previous portfolio-management sequence on covariance, multi-asset optimisation, Markowitz theory, and efficient frontiers.

Investigation / Activity

The result is one coherent derivatives lecture rather than a combined derivatives-and-portfolio lecture.

⋆ Core Final Synthesis: What Lecture 9 Adds To Risk Management I

Key Definition / Result

Lecture 9 shows that derivatives are not merely speculative products. They are contracts that reshape future payoffs.

- Forwards and futures lock future prices but create obligations.
- Futures can hedge risk, but margin and leverage can amplify losses.
- Options create asymmetric rights and require premiums.
- Put-call parity introduces replication and no-arbitrage thinking.
- Local structured products such as warrants and CBBCs embed derivative-style payoff rules that students must understand before use.

★ Core

Wrap-Up And Next Steps I

Technology Check: Moved To Tutorials / Labs

- More payoff and profit diagrams for forwards, futures, calls, and puts.
- Spreadsheet or R plotting of option payoff functions.
- More numerical examples on margin, leverage, and futures gains or losses.
- Optional no-arbitrage explanation of put–call parity.
- Optional self-directed exploration of Black–Scholes–Merton replication.

End Of Lecture 9